

Module & Submodule Reference

A complete, verified map of every functional module in the internal-audit platform and the submodules within it — with the API surface, primary domain aggregates, permissions, and the business rules that govern each area. Compiled directly from the source (controllers, domain aggregates, and Angular feature areas).

ASP.NET Core 10 · Angular 21 · Clean Architecture (Domain / Application / Infrastructure / API) · hand-rolled CQRS · EF Core 10 / SQL Server

How the platform is organised. AuditX follows the audit value chain end to end: it maintains the *audit universe* and a risk-based *annual plan*, authors reusable *checklist templates*, runs *audit engagements* and *fieldwork*, tracks *findings* to remediation, governs *controls*, *compliance* and *enterprise risk*, administers *sanctions*, and delivers *reporting*, *analytics* and *audit-committee* governance — all on a hardened, single-tenant on-premises stack.

Every module is a vertical slice (domain → application → infrastructure → API → Angular feature), guarded by fine-grained permissions, optimistic concurrency, and an append-only audit trail.

READING THE ENTRIES

- **API** — REST base route(s) under `/api/v1`.
- **Aggregate** — the primary DDD domain root.
- **Perms** — representative permission keys gating the area.
- **UI** — the Angular feature folder.
- Every endpoint returns an enveloped `{ data }`; mutations echo a `version` rowversion for concurrency.

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01 Identity & Access M1

API `/auth` `/users` `/roles` `/permissions` `/maker-checker` AGGREGATE User · Role · UserRole · MakerCheckerAction UI `features/auth` `features/admin/identity`

Authenticates staff against Active Directory (no passwords stored) and governs who can do what through AuditX-owned roles, scoped permissions, dual control, and time-bounded delegation.

Authentication & sessions POST <code>/auth/login</code> · GET <code>/auth/sso</code> · <code>/auth/session</code>	AD credentials validated by an LDAP bind, or Kerberos/IWA single sign-on; a Development provider covers local use. Sessions are short-lived JWTs in HttpOnly cookies (8h sliding / 24h absolute) with a Redis denylist for revocation; the session endpoint re-verifies AD account status on every call.
Users & provisioning <code>/users</code> · <code>/users/me</code> · <code>/users/directory</code>	Just-in-time provisioning creates users in an <i>awaiting-role</i> state; granting the first role activates them. Supports deactivation/reactivation, per-user annual capacity (person-days), notification preferences, and a lightweight <code>id→name</code> directory any signed-in user can read.
Roles & inheritance <code>/roles</code> (ManageRoles)	Roles are AuditX-owned and independent of AD groups, with an acyclic inheritance graph and permission keys validated against the catalogue. Four built-in roles (Auditor, Audit Manager, Auditee, Administrator) are immutable; the Administrator holds every permission. Permission changes are maker-checker gated.
Permission catalogue GET <code>/permissions/catalogue</code>	The full machine-readable catalogue of permission keys and their scopes, used to drive the role editor and to validate grants. Scoped permissions can be global or bound to a specific resource.
Maker-checker (dual control) <code>/maker-checker</code> · <code>gates</code>	High-risk actions are captured as pending actions with a serialized payload replayed on approval; a maker can never approve their own action, and rejections need a ≥20-char reason. Admin-configurable gates cover role-permission change, template publish, MAP approval, sanctions-grid edit, and config activation.
Delegation <code>/users/{id}/delegations</code> (Delegate)	Time-bounded delegation of a role from one user to another (modelled on the UserRole entity); the delegated authority only confers permission inside its window. An hourly background job expires elapsed delegations; either party can terminate early.

02 Audit Universe & Org Structure M2 · M3

API `/audit-universe` `/risk-dimensions` `/org-units` AGGREGATE `AuditableEntity` · `RiskDimension` · `OrgUnit` UI `features/audit-universe`

The catalogue of everything auditable and the organisational tree it rolls up to — the risk-based foundation the annual plan prioritises against.

Auditable entities <code>/audit-universe/entities</code>	Branches, processes, systems, vendors and products, with owner, org-unit, an acyclic parent hierarchy, and a last-audited stamp set when an audit completes. Each entity carries an <i>expected-audits-per-year</i> coverage target and supports CSV bulk import (soft-delete/archive on removal).
Risk scoring & dimensions <code>/entities/{id}/risk-scores</code> · <code>/risk-dimensions</code>	Entities are scored against a configurable set of weighted risk dimensions; the composite is a weight-normalised average over each dimension's min/max scale, with full score history. Scoring rejects incomplete/unknown/out-of-range inputs so prioritisation stays defensible.
Organisational units <code>/org-units</code> (<code>ManageUniverse</code>)	An acyclic division → department → branch/unit tree with unique codes; the roll-up dimension for org-based reporting. Units can be renamed, reparented, archived and restored.
Org-unit scorecards GET <code>/analytics/org-units</code>	Department / business-unit scorecards that aggregate audits and findings up the org-unit subtree, so a parent unit reflects the activity of all its children.

03 Annual Planning M3

API `/annual-plans` `/coverage-analytics` AGGREGATE `AnnualPlan` · `PlanItem` · `PlanItemEntityLink` UI `features/planning` , `features/coverage`

Builds the period audit plan, routes it through audit-committee approval, and locks it once approved so audits happen as planned unless a governed revision is made.

Plans & lifecycle Draft → Submitted → Approved → Closed	Period-scoped plans move Draft → Submitted → (Revisions Requested / Approved) → Closed; submit requires at least one item and raises an event for the committee. Editing is confined to Draft (and Revisions-Requested).
Multi-entity plan items POST <code>/{{id}}/items</code>	One plan item is a single schedule (audit type + window + lead) that can cover several entities; each entity gets its own link and its own independently-launched audit, with a rolled-up item status. Item dates must fall inside the plan period and stay ordered.
Committee routing <code>/submit</code> · <code>/decision</code> (<code>ACChair</code>)	The plan is submitted for approval; the committee chair records a decision (approve / request revisions / reject) with comments, persisted as an immutable decision record. Approval stamps the approval time.
Post-approval governance <code>/submit-revision</code>	An approved plan locks: a <i>minor</i> revision shifts item dates in place and stays approved, while a <i>material</i> revision re-opens the plan for committee re-approval. A bank setting can permit launching audits before approval.
Coverage analytics <code>/coverage-analytics</code> (<code>ViewCoverage</code>)	Not-audited-since, high-risk coverage gaps, and a coverage matrix over a window — the views that expose where the universe is under-covered relative to its risk and expected frequency.

04 Checklist Templates M2

API `/templates` `/rating-scales` `/response-option-sets` AGGREGATE `Template` · `TemplateVersion` · `RatingScale` · `ResponseOptionSet` UI `features/admin/templates`

Reusable, versioned checklists that seed an audit's fieldwork, plus the organisation-configurable scoring and conclusion vocabularies that back them.

Template authoring & publish <code>/templates</code> (<code>ManageTemplates</code>)	Draft authoring of sections and items, then publish (maker-checker gated) into an immutable version; a new draft can be spun from a published template, and templates can be cloned, archived and diffed across versions.
Sections & items <code>/{{id}}/items</code> · <code>/{{id}}/sections</code>	Ordered sections and items; each item declares a response type, reference notes, required flag, optional default-assignment rule, per-type config, an item risk rating (drives default exception severity), and an optional linked control. Response types are PassFailNa and YesNo (verdict) plus Text, Numeric, Date, Rating and MultipleChoice (value).
Rating scales <code>/rating-scales</code>	Reusable labelled 0–100 point sets referenced by Rating-type items; the point an auditor selects drives the item's post-response score.
Response option sets <code>/response-option-sets</code>	Organisation-defined conclusion options per verdict type — relabel and re-score the choices (e.g. Compliant / Partially Compliant / Non-Compliant / N-A). Each option carries its own score and semantics, and the engine derives a canonical Pass/Fail/N-A so scoring, findings and reports keep working; resettable to defaults.

05 Audit Engagements M4

API `/audits` AGGREGATE `Audit` · `AuditTeamMember` · `AuditSection` · `AuditChecklistItem` UI `features/audits`

The engagement itself — its lifecycle state machine, team, checklist, and the pre-audit and self-assessment modes that adapt how an audit is run.

Lifecycle Draft → Planned → In Progress → Under Review → Completed	Planning requires a checklist item and an auditor; review can auto-trigger when the last item is finalised; completion stamps the actual end date and updates the linked plan item and entity. Return-to-progress, reopen and cancel (≥20-char reason) round out the machine; transitions are resource-scoped in the handler.
Team management <code>/{{id}}/team</code> · <code>/transfer-lead</code>	Lead, auditee and auditor membership with lead transfer; the sole lead or auditee can't simply be removed, and a member's items must be reassigned before they leave.

Checklist authoring

`/{{id}}/checklist/items` · `/sections`

First-class item and section CRUD with reorder and drag-drop *arrange* (cross-section moves that auto-create sections). Items are copied from the template version at creation, so a running audit is decoupled from later template edits.

Item assignment

`/{{id}}/items/{{itemId}}/assignment` · `bulk-reassign`

Assign or clear the owner of a checklist item (must be an active team member), individually or in bulk.

Kickoff meeting

`POST /{{id}}/kickoff`

Schedule or reschedule a pre-audit kickoff meeting (time, location/join link, agenda); scheduling notifies the auditee in advance through the notification pipeline.

Self-assessment mode

`POST /audits/self-assessment`
(`RunSelfAssessment`)

The area owner assesses their own area — the same person is both lead and auditee, no independent auditor is required, and only they may respond — reusing the full checklist/response/exception engine from a published template.

06 Execution & Fieldwork M5

API `/audits/{{id}}/...responses` `/evidence` `/evidence-requests` `/time-entries` `/procedures`

AGGREGATE `ChecklistResponse` · `EvidenceFile` · `EvidenceRequest` · `TimeEntry` · `AuditProcedure` UI `features/audits/audit-execution`

The working-paper layer: recording conclusions and their evidence, and capturing the effort and procedures behind the fieldwork.

Checklist responses

`POST /{{id}}/items/{{itemId}}/responses`

One create-or-update response per item: verdict (or custom option), captured value, comment, *observation* and *recommendation*, with a 0–100 post-response score and full before/after edit history. Fail/N-A always require a comment; a material edit re-computes the score.

Evidence

`/{{id}}/responses/{{rid}}/evidence` · `/evidence/flagged`

Uploads are magic-byte + declared-MIME validated and SHA-256 hashed, then re-verified on every download — a mismatch flags the file and blocks reads until an admin unflags it. Per-file and per-audit size caps apply; deletes are soft with a mandatory reason.

Evidence requests

`/{{id}}/evidence-requests`
(`RespondItem`)

The expected/requested side of evidence — request a document (type, due date), then mark received or waive (with reason); overdue-but-still-requested items drive missing-evidence reporting.

Time tracking

`/{{id}}/time-entries` · `/summary`
(`LogTime`)

Logged effort by category (Fieldwork, Review, Reporting, Planning, Administration) with amendments; the sum is the "actual" side of budget-vs-actual against the audit's budgeted hours, grouped by auditor for utilisation.

Procedures

`/{{id}}/procedures` (`RespondItem`)

Typed fieldwork procedures — sampling, interview and walkthrough — with a coherent sampling chain (population ≥ sample size ≥ items tested ≥ exceptions found) so the analytics error-rate never counts orphan exceptions.

Review worklists

`/{{id}}/review/fail-without-exception` · `/summary`

Surfaces failed items that lack a raised exception and lets a manager record a justification that a Fail needs no exception, plus a review-summary roll-up before sign-off.

07 Findings & Exceptions M6

API `/exceptions` `/exception-raising-rules` `/root-cause-gaps` AGGREGATE `AuditException` · `MapAction` · `FindingVerification` · `RootCauseGap`

UI `features/exceptions`

The finding lifecycle from being raised through management action, closure and follow-up — plus the systemic root-cause register that ties recurring findings together.

Exceptions & the finding register

`/exceptions` (`ViewExceptions`)

Findings carry severity (defaulted from the item's risk rating), owner, financial impact, root-cause and non-conformance categories, and recurrence links; a cross-audit tracker and hash-stamped CSV export span the whole register. Severity/owner changes need reasons and are blocked once terminal.

Status lifecycle

`Open` → `MapSubmitted` → `MapApproved` → `PendingClosure` → `Closed`

Exactly seven states (plus Cancelled/Rejected edges). A Critical finding's closure sets a CIA-hold flag inside `PendingClosure` and stays there until the Chief Internal Auditor countersigns; reopening a closed finding increments a reopen count.

Management action plans (MAP)

`/{{id}}/map` · `/map/approve` · `/map/mark-complete`

The auditee submits remediation actions (owner, target date, expected evidence), which are approved (maker-checker gated) or rejected; completing every action moves the finding to pending-closure. Actions accept their own evidence uploads.

Verification & links

`/verifications` · `/controls` · `/regulations`

Post-closure follow-up verifications (passed / partially / failed) provide durable assurance, and findings link many-to-many to the controls and regulations they implicate — powering compliance-by-regulation and control-failure analytics.

Exception-raising rules

`/exception-raising-rules`

Bank-configurable rules that *extend* when a response is eligible to raise a finding — on an N/A verdict, or a score at/below a threshold — over and above the always-eligible Fail verdict.

Root-cause gaps & remediation plans

`/root-cause-gaps`

Systemic weaknesses tracked as first-class open→closed items that link the contributing findings, each with a remediation plan of corrective actions that carry their own owner, due date and Open→Completed state.

08 Controls & Compliance

API `/controls` `/regulations` AGGREGATE `Control` · `ControlTest` · `ControlRiskLink` · `Regulation` UI `features/controls` `features/compliance`

The internal-control register and its effectiveness testing, wired to the risks it mitigates, the items that test it, and the regulations it satisfies.

Control register <code>/controls</code> (ViewControls / ManageControls)	Controls with an immutable code, type (Preventive / Detective / Corrective / Directive), frequency, owner and covered entity; effectiveness starts Not-Tested and any assessed effectiveness requires a last-tested date. Controls can be activated or retired.
Effectiveness testing <code>/controls/{id}/tests</code>	Append-only tests — driven by a checklist item during an audit or recorded ad-hoc — set the control's current effectiveness (Effective / Partially / Ineffective) and last-tested date; the latest test reflects onto the control.
Risk & item linkage <code>/controls/{id}/risks</code> · <code>item#control</code>	Controls link many-to-many to enterprise risks (neither register owns the other), and checklist items reference the control they test so that raising a finding auto-links it to that control.
Regulations register <code>/regulations</code>	Regulatory obligations with an immutable code, issuing authority and category; findings link to regulations to power the compliance-by-regulation report.

09 Enterprise Risk

API `/risks` AGGREGATE Risk UI `features/risks`

A standalone enterprise-risk register with inherent/residual likelihood-and-impact scoring and an ISO-31000 treatment lifecycle — distinct from the universe's audit-prioritisation dimensions.

Risk register & scoring <code>/risks</code> (ViewRisk / ManageRisk)	Each risk scores likelihood × impact on a 1–5 scale; the current score is the residual where assessed, else inherent, driving a Low/Medium/High/Critical band and the 5×5 heatmap. Residual may not exceed inherent, and a complete residual pair auto-advances the risk to Assessed.
Treatment & lifecycle <code>/risks/{id}/transition</code>	Status flows Open → Assessed → Mitigating → Monitoring → Closed (terminal, with a rationale); the treatment strategy is Accept / Mitigate / Transfer / Avoid, and an optional entity link enables org-unit roll-ups.

10 Sanctions & Discipline M7

API `/sanctions/cases` `/sanctions/grid` AGGREGATE SanctionsCase · SanctionsAppeal · SanctionsGridVersion UI `features/admin/sanctions`

The disciplinary workflow triggered from a finding and run through recommendation, HR, the disciplinary committee and appeal — deliberately decoupled from the finding it came from.

Case lifecycle recommendation → HR → DC → appeal → closed	A case is triggered off an exception (copying its category/severity at trigger), a recommendation is drafted against the grid and submitted, HR records an outcome or refers to the disciplinary committee, the committee decides (with a voting record), and either party may appeal. The case never reads or writes the originating finding's state.
Grid consult & deviation <code>/cases/{id}/recommendation</code>	The recommendation consults the active sanctions grid for the recommended range; an out-of-range recommendation requires a documented deviation reason, keeping outcomes consistent and defensible.
Appeals <code>/cases/{id}/appeals</code> · <code>/appeals/{id}/decision</code>	An appeal is its own record routed to the configured appeals authority (default CIA) and decided once; the outcome propagates onto the case trail. Segregation of duties keeps closure a write-permission action, not a viewer's.
Sanctions grid <code>/sanctions/grid</code> (ViewGrid / ManageGrid)	The versioned offence-matrix mapping (category severity recurrence) → recommended range; exactly one version is active bank-wide, and activating a new version is maker-checker dual-controlled with a reason.
Case dossier POST <code>/cases/{id}/dossier</code>	Generates a hash-stamped case dossier document (SHA-256 header) capturing the case record for HR/committee review.

11 Reporting M8

API `/reports` `/report-templates` `/report-schedules` `/shared-links` AGGREGATE Report · ReportDistribution · ReportTemplate · SharedLink UI `features/reports`

Tamper-evident report generation, distribution with delivery confirmation, and the templates, schedules and shareable links around it.

Report generation <code>/audits/{id}/reports</code> · <code>/reports/standalone</code>	Per-audit engagement reports and standalone cross-audit reports (executive summary, plan status, KPI pack) render asynchronously and record a canonical HTML SHA-256; the template definition is snapshotted at start so historical reports render as issued. Formats produced: HTML, PDF, CSV, XLSX (DOCX on request); completed reports are immutable.
Integrity verification <code>/reports/{id}/download</code> · <code>/verify-hash</code>	Every download re-verifies the SHA-256 and stamps it on the response; a recompute-and-compare endpoint and a hash-mismatch alert event provide tamper evidence without mutating content.
Distribution & delivery <code>/reports/{id}/distribute</code> · <code>/distributions/{id}/outcome</code>	Distribute to users, emails or roles and record a per-recipient delivery outcome (delivered / bounced), with a full distribution log.
Templates & schedules <code>/report-templates</code> · <code>/report-schedules</code>	Bank-configurable report layouts (versioned, activated with a reason) and recurring schedules that auto-generate a standalone report on a cadence and email recipients, executed by a background runner as the System actor.
Shared links <code>/reports/{id}/share</code> · <code>/shared-links/{slug}</code>	Opaque, optionally-expiring links that are a reference — not an access grant: resolving a slug returns only the target descriptor, and the viewer must still authenticate and hold the report's permission. Links are revocable and retained for the trail.

12 Analytics M9

API [/analytics](#) [/dashboards](#) AGGREGATE KPI query services · Dashboard · widgets · snapshot fact table UI [features/analytics](#) [features/dashboard](#)

A read-only KPI suite over the whole platform, with trend history and configurable dashboards — and a hard privacy rule that no analytics projection exposes a sanctions subject's identity.

Function & portfolio KPIs

[function-performance](#) · [exception-portfolio](#) · [material-findings](#) · [plan-status](#)

Headline health of the audit function, the finding portfolio, material findings and annual-plan status (the source for the plan Gantt) — the bank-wide read-only indicators for management.

Performance scorecards

[performance-scorecards](#) · [auditor-throughput](#)
([PerformanceAnalyticsView](#))

Per-auditor and team throughput scorecards, gated by a dedicated permission and suppressing the caller's own row unless they are the CIA — so performance data is seen appropriately.

Coverage, workload & risk

[coverage](#) · [auditor-workload](#) · [risk-heatmap](#) · [control-effectiveness](#) · [compliance-by-regulation](#)

Coverage heatmap, workload-versus-capacity, the 5×5 risk heatmap, control-effectiveness mix and compliance-by-regulation — plus budget-vs-actual, utilisation, finding follow-up, procedure error-rates and evidence turnaround.

Recurrence detection

[recurrence-clusters](#)

A daily scan clusters recurring control weaknesses across the finding history and raises an alert when a new cluster crosses the recurrence threshold.

Trends & comparison

[trend](#) · [metric-comparison](#) · [snapshots/capture](#)

Time-series from a daily KPI snapshot fact table, with month/quarter/year-over-year deltas and a linear forecast; snapshots can be force-captured for the day.

Configurable dashboards

[/dashboards](#) ([ConfigureDashboards](#))

Widget-composed dashboards with an optional per-dashboard permission enforced in-handler, so a caller never even learns a gated dashboard (e.g. sanctions consistency) exists.

13 Audit Committee M13

API [/ac-packs](#) [/ac-action-items](#) [/ac-comments](#) [/ac-dashboard](#) AGGREGATE [AcPack](#) · [AcActionItem](#) · [AcComment](#) · [FindingVisibilityRestriction](#) UI [features/ac](#)

The committee-facing layer: assembling and distributing board packs, tracking committee actions, and the collaboration and visibility controls around them.

AC packs

[/ac-packs](#) ([generate](#) / [approve](#) / [distribute](#))

A period pack snapshots the analytics into immutable content and a canonical SHA-256; the CIA adds supplementary narrative and single-actor approves it (which re-seals the hash to include the text) before distribution. Members only see approved/distributed packs.

Action items

[/ac-action-items](#)

Committee-raised follow-ups that flow Open → In-Progress → Closed (closure requires a written response) → Acknowledged by the chair (terminal).

Dashboard & comments

[/ac-dashboard](#) · [/ac-comments](#)

A read-only committee dashboard of live aggregates (never subject identity) and polymorphic comment threads on pack contents and other targets.

Finding visibility

[/findings/{type}/{id}/restrict-visibility](#) (CIA)

A per-finding allow-list, set by the CIA, applied per-requester at read time so sensitive findings are seen only by named people.

14 Notifications M10

API [/notification-rules](#) [/notification-templates](#) [/notification-dispatches](#) AGGREGATE [NotificationRule](#) · [NotificationTemplate](#) · [NotificationDispatch](#)
UI [features/admin/notifications](#)

A durable, data-driven pipeline that turns domain events into templated messages over email, SMS and Microsoft Teams, with retries and a dead-letter queue.

Rules & event catalogue

[/notification-rules](#) · [/admin/events/catalogue](#)

Each rule maps a subscribable domain event to recipients (role, named users, or a user id derived from the event payload) and channels; rules can be previewed for their resolved recipients and rendered template before going live.

Templates

[/notification-templates](#)

Per-channel message templates with bank overrides on top of seeded system defaults.

Channels

[Email](#) · [SMS](#) · [Teams](#)

Email over SMTP, SMS over the bank gateway, and Microsoft Teams via an incoming webhook (delivered once per event/rule as a channel broadcast rather than per recipient), routed centrally.

Dispatch, retry & dead-letter

[/notification-dispatches](#) · [/dead-letter](#) ([AdminOps](#))

Every send is a tracked dispatch with backoff retries (1m / 5m / 15m / 1h) before dead-lettering; the dead-letter queue and manual re-queue are ops-only.

15 Platform & Administration M11 · M14

API [/admin](#) [/audit-trail](#) [/configurations](#) [/reference-data](#) [/integrations](#) [/engagements](#)
AGGREGATE [BankSettings](#) · [Configuration](#) · [Integration](#) · [SavedView](#) · [AuditTrail](#) UI [features/admin/*](#) [features/engagement-lifecycle](#)

The cross-cutting administration and platform services every other module relies on.

Bank settings & ops <code>/admin (bank-settings, resource-limits, releases, restore, support-channel)</code>	Bank identity, branding, retention and session settings; resource limits; bulk user operations; signed offline release installs; backup restore-drills and approval-gated object restore; and system-health.
Audit trail <code>/audit-trail (ViewAuditTrail / ExportAuditTrail)</code>	An append-only activity log with before/after snapshots, enforced at the database by a trigger; queryable by actor/type/object/date with facets, per-object history, and a tamper-evident hash-stamped export.
Configuration & reference data <code>/configurations · /reference-data</code>	A generic versioned config store (draft → activate → rollback, one active version per domain, maker-checker-gated activation) and the managed reference-data lists (audit types, categories, document types) that back every dropdown.
Integrations & webhooks <code>/integrations · /webhook-subscriptions</code>	Configurable outbound integrations with health checks and fallbacks, and HMAC-signed webhook subscriptions to a curated event catalogue with a delivery log and retry.
Saved views & search <code>/saved-views · /search</code>	Per-user (optionally shared) named filter sets that pre-fill a screen's filters, and a cross-module global search that only returns hits the caller is permitted to see.
Engagement lifecycle <code>/engagements (ViewEngagementLifecycle)</code>	A read-only portfolio board and per-engagement journey (setup → fieldwork → review → remediation → reporting → closed) with a deterministic <i>next-best-action</i> engine that only suggests actions the user is permitted and positioned to take.
Branding & health <code>/branding (anon) · /admin/health (anon)</code>	A public branding endpoint so the login shell can theme itself before a session exists, and an anonymous liveness/readiness probe checking the database and Redis.